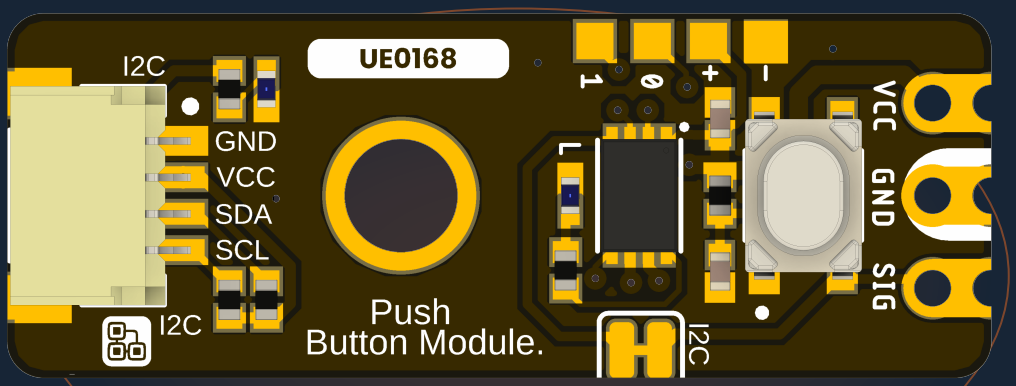
PRODUCT REFERENCE MANUAL

UNIT ELECTRONICS · DEVLAB ECOSYSTEM

DevLab I2C Push Button Module

MPN: UE0168

Version 1.1.0



I2C push-button module for digital user input.

PUSH BUTTON | I2C | DEVLAB ECOSYSTEM

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Description

The DevLab I2C Push Button Module (MPN **UE0168**) is an I2C push button input module in the UNIT Electronics DevLab family. The product-specific board views and any released drawings are collected under `hardware/`.

Applications

- Push button input in an I2C-based prototype.
- Evaluation of the module with a compatible host controller.

Available Documentation

A product-specific schematic was not present in the selected source package. See Chapter 8 for the available source files. Confirm the electrical interface against the product-specific pinout or schematic before connecting the board.

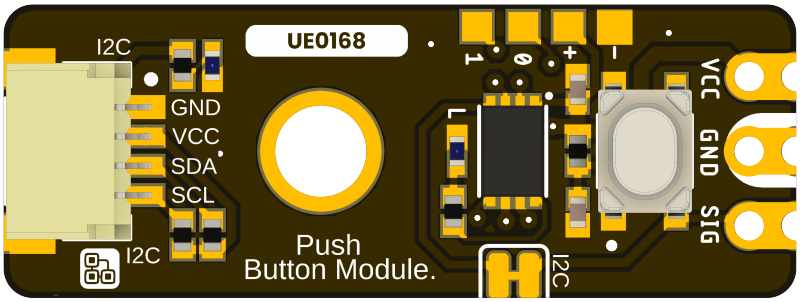
1 The Board

1.1 Board Identification

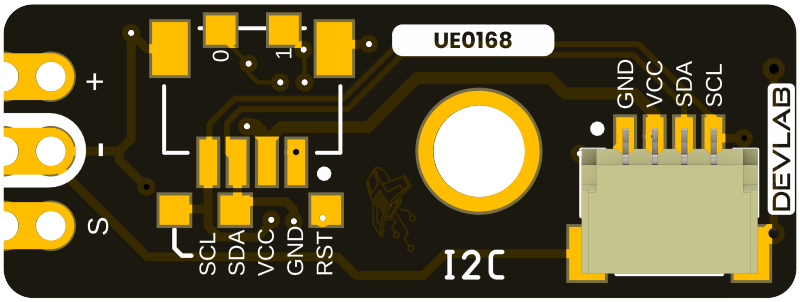
Item	Identification
Product	DevLab I2C Push Button Module
Manufacturer Part Number (MPN)	UE0168
Function	Push button input

1.2 Board Views

The following views come from the selected UE0168 source package. They show component placement and board orientation; use a dimensioned drawing for mechanical tolerances.



DevLab I2C Push Button Module top view.



DevLab I2C Push Button Module bottom view.

2 Ratings

The selected UE0168 source package does not establish a complete set of electrical ratings in this reference. Do not use values from another DevLab module as substitutes.

2.1 Electrical Limits

Supply range, I/O voltage limits, and current consumption require confirmation from the product-specific schematic and component data before integration.

2.2 Interface Characteristics

Confirm the I2C address, bus speed, pull-up arrangement, and connector orientation with the released UE0168 pinout or firmware documentation. An I2C scan only identifies responding addresses; it does not verify the device protocol.

3 Functional Overview

The DevLab I2C Push Button Module provides push button input for an I2C-based system. The supplied board views identify the physical module, while the product-specific schematic and firmware documentation are needed to establish its internal signal path and command set.

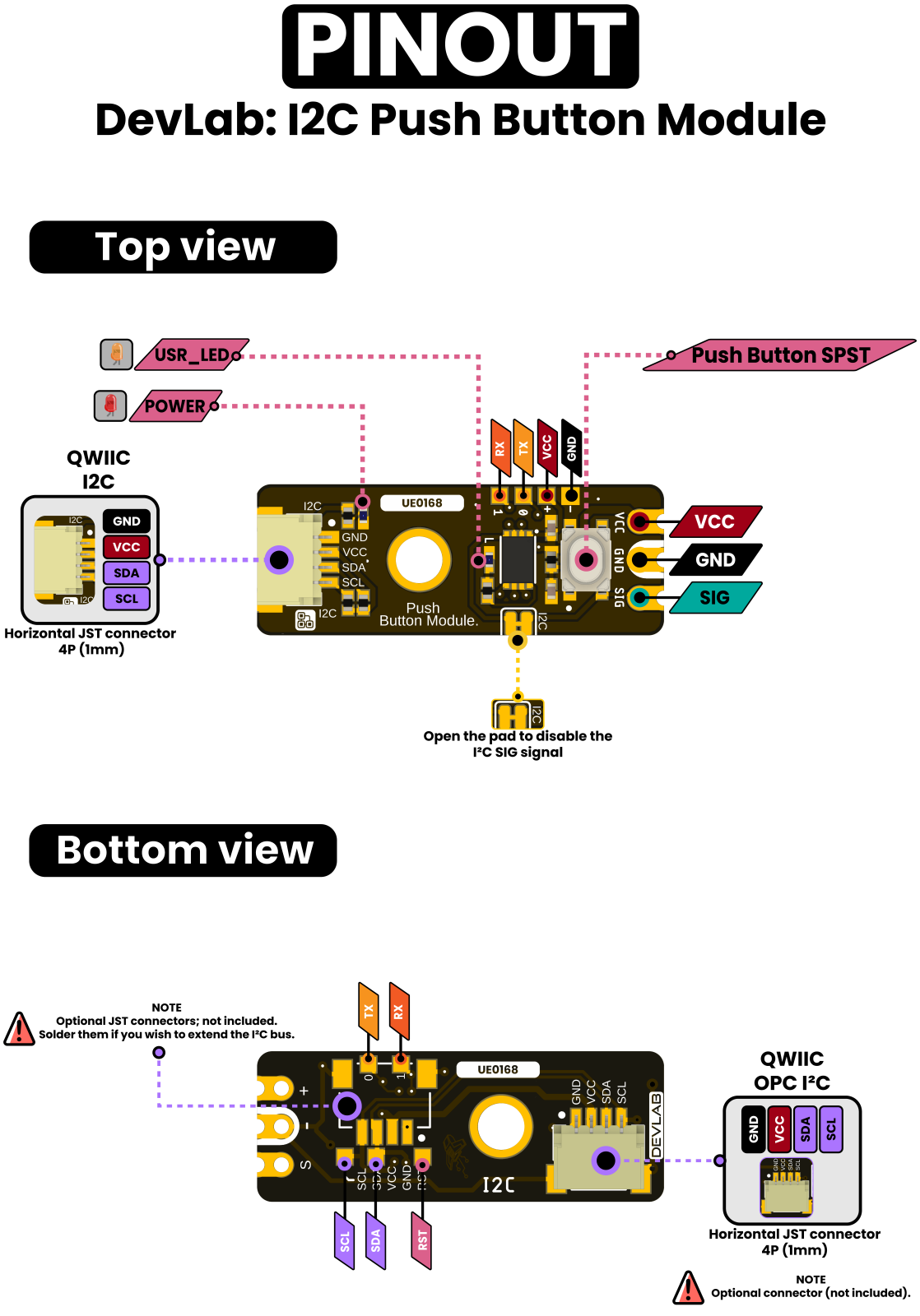
3.1 Integration Boundary

Use the available UE0168 pinout, schematic, and firmware release to determine power, connector mapping, address, and commands. Until these are confirmed, treat the generic I2C scanner in `software/` as a bus diagnostic only.

4 Connectors & Pinouts

The released V1.0 pinout for the UE0168 Push Button Module shows the top and bottom board views. Check the view and connector orientation before wiring.

4.1 Released Pinout



UE0168 Push Button Module pinout, top and bottom views.

Figure 4.1 — UE0168 pinout, English version. The Spanish version is linked in Section 8.

4.2 Connection Groups

Group	Labels in the released drawing	Purpose
Qwiic I2C connector	GND, VCC, SDA, SCL	Power and I2C bus connections
Direct contacts	VCC, GND, SIG	Power and direct signal access
Service pads	TX, RX, RST	Pads identified in the drawing; consult the hardware design before use

The drawing also marks two optional 4-pin, 1 mm JST connector positions. Its solder-link note says opening the marked pad disables the I2C SIG signal. The pinout does not establish electrical limits or connector part numbers.

5 Board Operation

5.1 Before Connecting

Check the product-specific UE0168 pinout or schematic for power, ground, SDA, and SCL. The board views alone do not establish voltage limits or connector numbering.

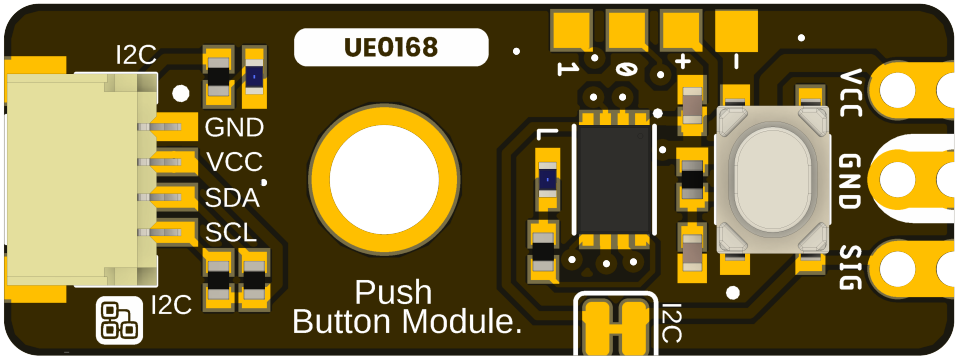
5.2 I2C Bus Check

After confirming the wiring and supply limits, use the generic Arduino or MicroPython I2C scanner in `software/examples/i2c/`. Set the host SDA and SCL pins for your controller. A responding address confirms only that a device acknowledged on the bus; it does not identify the module or validate its measurement protocol.

6 Mechanical Information

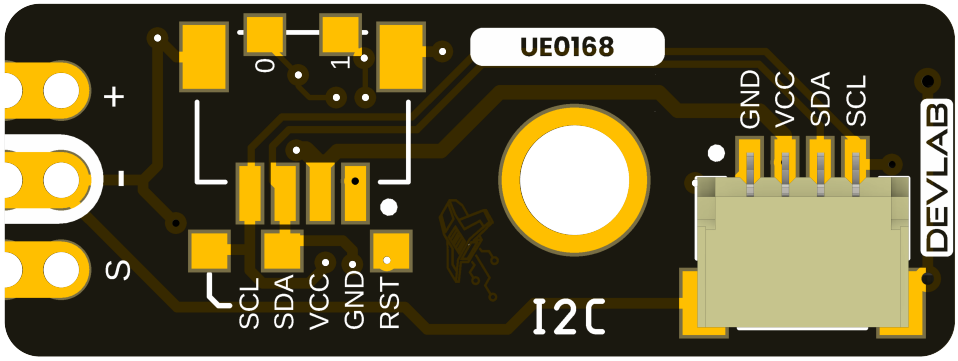
The supplied board views identify the component and underside layouts of the UE0168 I2C Push Button Module. Use the labeled drawings for fit checks; do not infer dimensions by scaling a render.

6.1 Top View



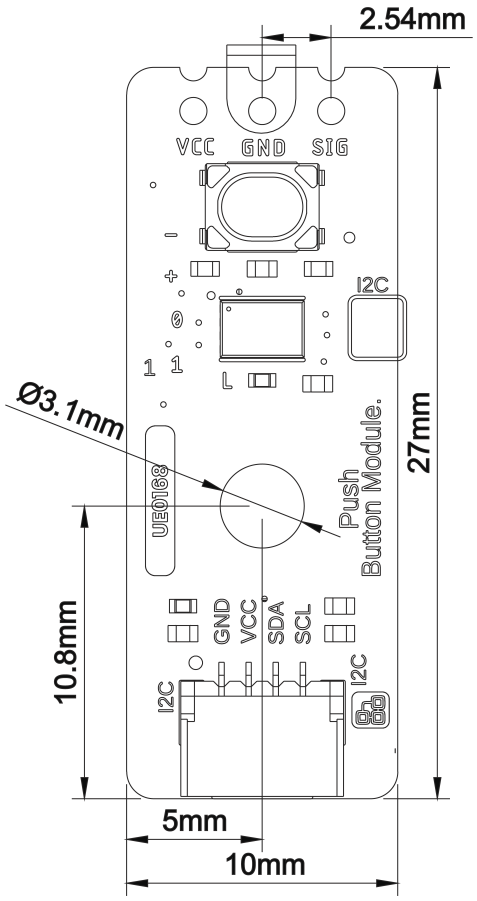
UE0168 I2C Push Button Module top view.

6.2 Bottom View



UE0168 I2C Push Button Module bottom view.

6.3 Board Dimensions



Board Dimensions

UE0168 I2C Push Button Module dimension drawing.

Use the dimensions printed on the supplied drawing. PCB thickness, component heights, connector clearance, and manufacturing tolerances require separate verification.

7 Company Information

Item	Information
Company / brand	UNIT Electronics
Board ecosystem	DevLab
Product family	Atom
Website	UNIT Electronics
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The product repository and controlled technical references are listed in the next chapter.

8 Reference Documentation

Reference	Location
Product repository	UE0168 Push Button Module
Pinout V1.0, English	PDF · PNG
Pinout V1.0, Spanish	PDF · PNG
Top view	3D board view
Bottom view	3D board view
Board dimensions	Dimension drawing
Board topology	Topology drawing
CAD model	STEP

The released pinout documents the visible labels and connector orientation. It does not by itself establish supply limits, bus timing, or mating part numbers.

9 Appendix

9.1 Schematic Status

A product-specific schematic was not included in the selected source package.

9.2 Document Control

Item	Value
Product	DevLab I2C Push Button Module
MPN	UE0168
Source	UNIT Electronics selected product source package

9.3 Data Requiring Confirmation

Electrical limits, I2C commands, and mechanical tolerances must be verified against product-specific released documentation.